FEDERAL SPECIFICATION

CANNULA, EYE, CYCLODIALYSIS (CASTROVIEJO), 10 mm.

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one size of Castroviejo cyclodialysis eye cannula.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

GG–I–526—Instruments, Dental and Surgical; General Specifications.

QQ–B–626—Brass, Leaded and Non-Leaded; Rods, Shapes, Forging and Flat Products with Finished Edges (Bar, Flat Wire and Strips).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D .C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–12902—Packing of Surgical, Dental and Veterinary Instruments, Diagnostic Equipment, and Related Items.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

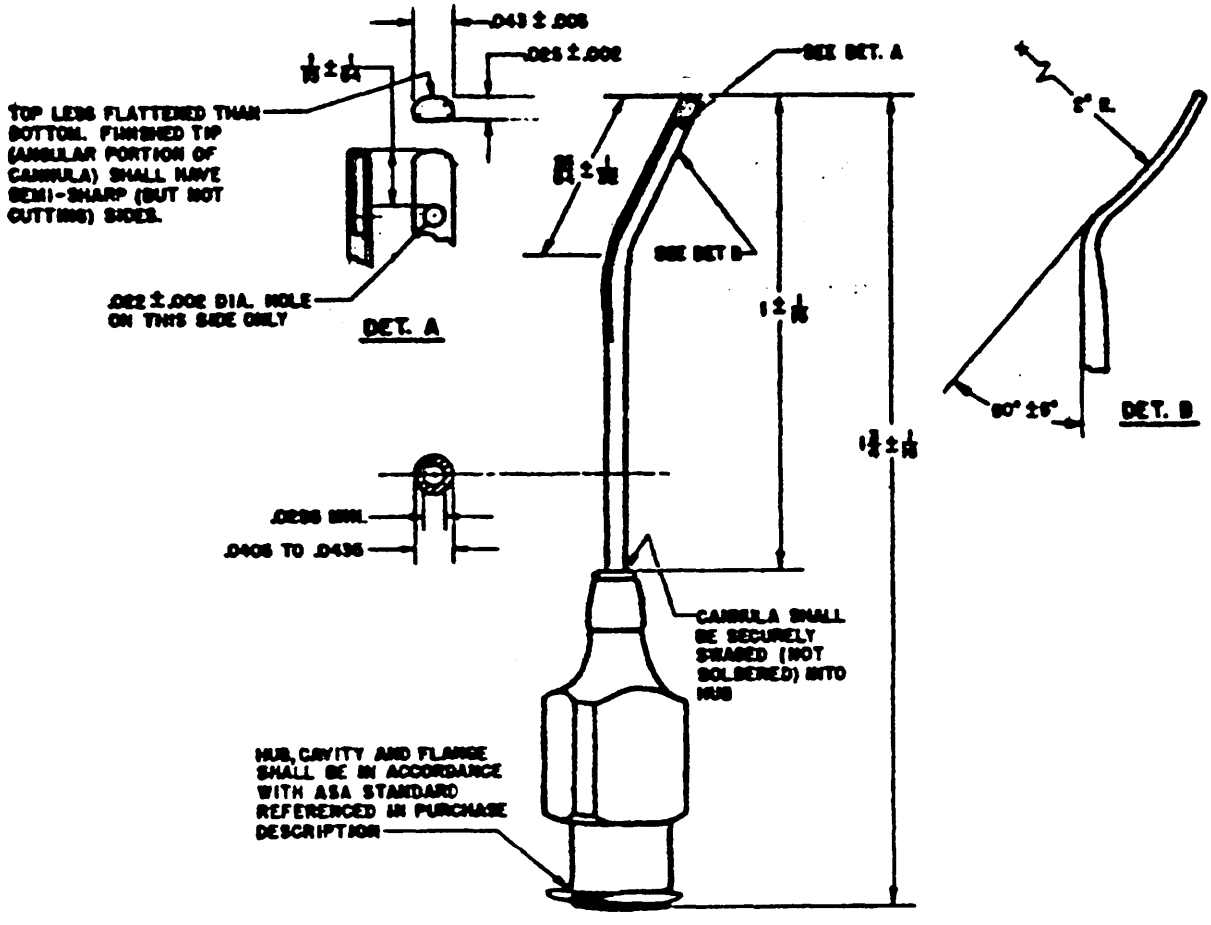
(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Standards Association, Inc. Standard:

Z70.1–1955 (UDC615.47) — Dimensions of Glass and Metal Luer Tapers for Medical Applications.

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NOTE

1. REMOVE BURRS.
2. DIMENSIONS ARE IN INCHES.

FIG. 1

(Application for copies should be addressed to the American Standards Association, Incorporated, 70 East Forty-fifth Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Cannula proper.* The cannula proper shall be fabricated from class 2, composition G steel conforming to the requirements of Federal Specification GG–I–526, except that the hardness requirement and the micrometallographic examination shall not apply. The cannula on the finished instrument shall be malleable.

3.1.2 *Hub.* The hub shall be fabricated from brass conforming to the requirements of Federal Specification QQ–B–626 for composition 22. The hub shall be nickel or chromium plated, and polished, and shall comply with the boil-test re-

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quirements of Federal Specification GG–I–526 for class 4, composition N material.

3.2 Style, design, and dimensions. The eye cannula shall be of the Castroviejo design, 10 mm. size. The instrument shall be suitable for use in performing a cyclodialysis in glaucoma and also in irrigating the anterior chamber of the eye. The instrument shall be in accordance with figure 1. The finished instrument shall provide for clear passage of fluids. The female Luer lock connection (i.e., the hub, its cavity, and flange) shall be in accordance with figure 3 of American Standards Association (ASA) Standard Z70.1–1955.

3.2.1 *Hub cavity.*

3.2.1.1 *Diameter and depth.* The diameter and depth of the hub cavity shall be inspected as specified in 4.4.1. Eye cannulae with hubs which do not fit gages within the limits as specified in 4.4.1 shall be considered not acceptable. In addition, when the hub is fitted onto gages as specified in 4.4.1, there shall be no movement of the hub on the gage when the cannula proper is pressed lightly from side to side in all directions.

3.2.1.2 *Freedom from foreign matter.* The hub cavity shall be tested for freedom from foreign matter as specified in 4.4.2, and shall show complete freedom from metal chips and other foreign matter.

3.2.1.3 *Fluid tightness.* The hub shall be tested for fluid tightness as specified in 4.4.3, and the connection of the cannula proper with the hub and the Luer connection of the hub to the plug gage shall show leakage not exceeding the formation of a falling drop when pressure is applied for 10 seconds.

3.3 Finish. The eye cannulae shall have a mirror finish conforming to the requirements of Federal Specification GG–I–526.

3.4 Identification marking. The cannulae of domestic manufacture shall be marked with the name or registered trademark of the manufacturer. Instruments of foreign manufacture shall be marked with the name of the country of origin, together with the name or registered trademark of the contractor.

3.5 Workmanship. Workmanship shall be first class throughout. Eye cannulae shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one cannula (end item).

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4.2.2 *Sampling for examination.* A random sample of cannula shall be selected from each lot offered to the Government in accordance with Military Standard MIL–STD–105. Examination for dimensions shall be in accordance with Military Standard MIL–STD–105, and as indicated in table I.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, and as indicated in table I.

4.2.4 *Raw materials.* Four ounces of each lot of each metal used in the construction of the cannulae shall be selected for chemical analysis.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The cannulae shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.

4.4.1 *Diameter and depth of hub cavity.* Hub cavity of the finished instrument shall be tested on the male Luer taper gage in accordance with plug gage of figure 5 of ASA Standard Z70.1–1955. This gage shall be as shown in the ASA Standard, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test as specified in 4.4.3. The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figure 5 of ASA Standard Z70.1–1955. The hub shall fit the plug gage within the "GO"-"NO GO" limits of the feeler gage as shown on figures 5 and 6 of the ASA Standard.

4.4.2 *Freedom from foreign matter.* The hub cavity of the finished instrument shall be tested for freedom from foreign matter by visual examination with a magnifier and good illumination, and also show complete absence of metal chips and other foreign matter upon visual examination under a magnifier of the cotton fibers of a cotton swab (tightly wrapped on the end of small diameter wooden applicator) which has been vigorously turned down into the bottom of the hub cavity.

4.4.3 *Fluid tightness.* With the cannula in a horizontal position, the hub attached by means of plug gage of figure 5 of ASA Standard Z70.1–1955 (except that gage shall have a fluid path) to a water pressure source having a pressure gage, the water shall be run through the cannula so as to fill any hub cavity space and the lumen, the lumen then sealed by a piece of solid rubber slipped onto the cannula point and

TABLE I.—*Sampling criteria*

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Cannula (end item)	Diameter and depth	3.2.1.1	4.4.1	1—cannula	II	1.0
	Freedom from foreign matter	3.2.1.2	4.4.2	1—cannula	L3*	**
	Fluid tightness	3.2.1.3	4.4.3	1—cannula	L7	1.5
	Clear passage of fluids	3.2	....	1—cannula	L7	1.0
	Dimensions	3.2	....	1—cannula	L3	4.0
Cannula proper (tube)	Boil test	GG–I–526	GG–I–526	1—cannula	L3*	**
Hub	Boil test	3.1.2	GG–I–526	1—cannula	L3*	**

* A minimum of three units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

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a pressure of 45 pounds per square inch applied for 10 seconds.

5. PREPARATION FOR DELIVERY

For civil agency procurement, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging and packing.

5.1.1 *Unit of issue.* One cannula, as specified, constitutes one unit of issue.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table II shall be packaged in each unit, intermediate package, and exterior container, as applicable, for the required level of protection specified in the procurement document.

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

TABLE II.—*Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	12 units	576 units

5.1.3 *Level A.* Packaging and packing shall conform to the requirements of Military Specification MIL-P-12902, Procedure Code No. 35.

5.1.4 *Level B.* Packaging and packing shall conform to the requirements of Military Specification MIL-P-12902, Procedure Code No. 21.

5.1.5 *Level C.* The subject commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking—nonmilitary.

5.2.1 Unless otherwise specified in the contract or order, unit packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3 Marking—military.

5.3.1 *Interior packages.* Marking shall conform to the requirements of Military Specification MIL-P-12902, except that the ASMPA stock number will not be required.

5.3.2 *Exterior container.* Exterior container shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1).

(1) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined the cannulae will be held in covered storage for not more than 12 months from date of initial packaging.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

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6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Surgical instruments, not otherwise indexed by name.

Carload minimum weight, 10,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Surgical instruments, not otherwise indexed.

Truckload minimum weight 10,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—M

Navy—Md

Air Force—AFCSC

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